



LSM-1003 Applied Mathematics

CLO 1.4 Summarize Data Using Measures of Central Tendency such as Mean, Median and Mode using Raw Data & Frequency Distribution

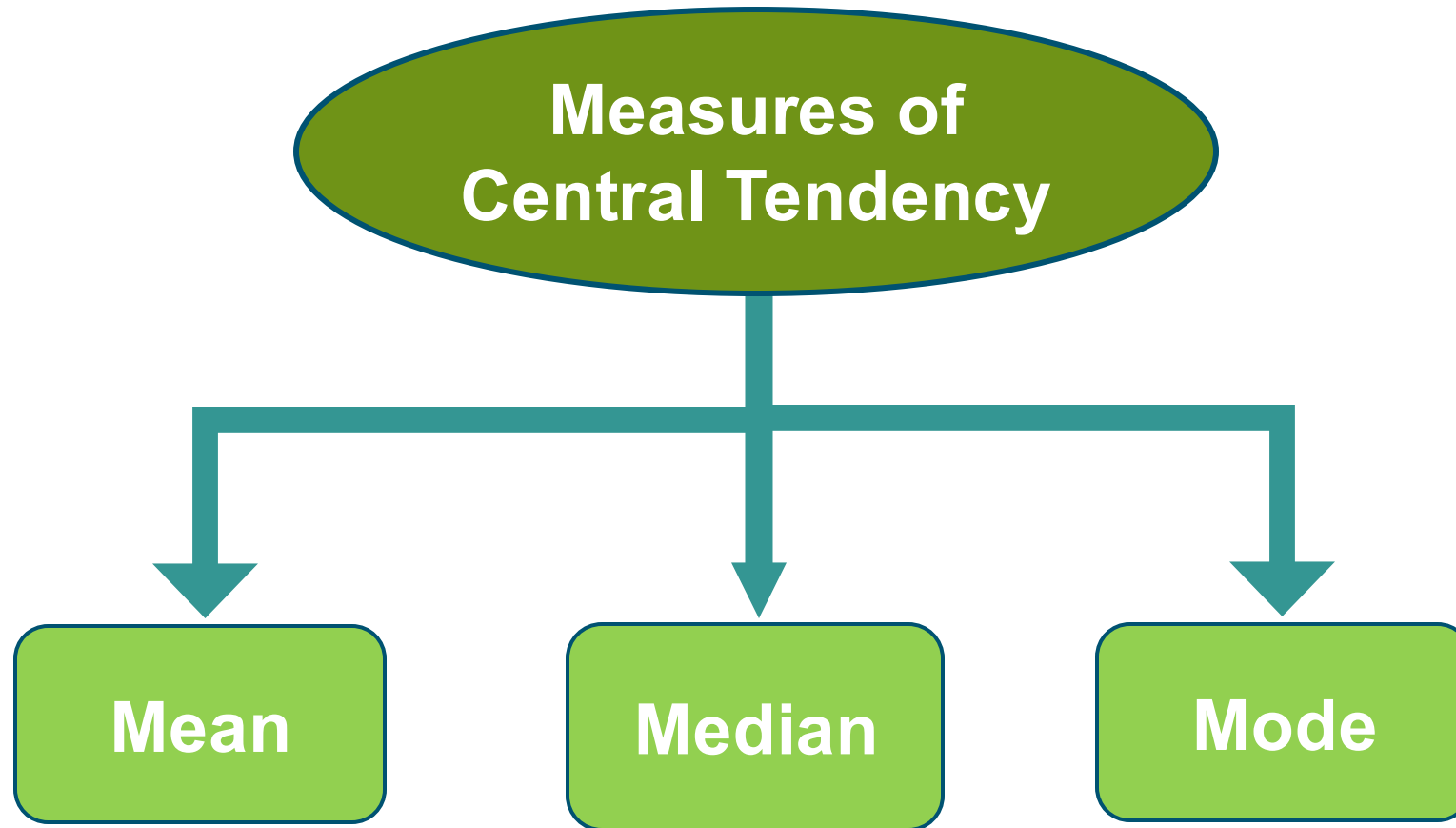


CLO 1.4

Summarize Data Using Measures of Central Tendency such as Mean, Median and Mode using Raw Data & Frequency Distribution

BUS 1023: Applied Mathematics for Business

Monday, September 21, 2026





What is the MEAN?



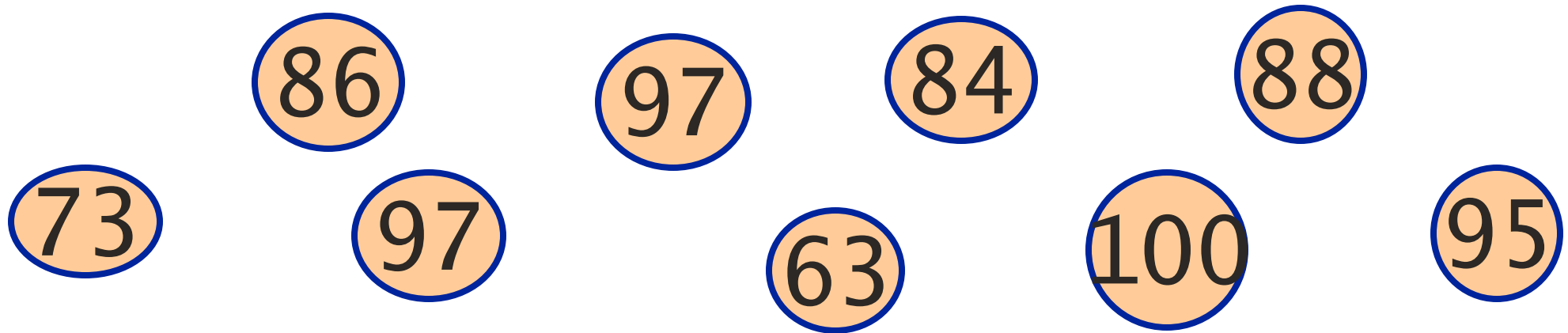
*The mean is the **numerical average** of the data set.*

$$\text{Mean} = \frac{\text{Total Value in the Set}}{\text{Number of Values}}$$

$$\text{Mean} = \frac{\sum X}{N}$$



These are Maha's Math's test scores.



Find the Mean.





Lets find Maha's MEAN Math's test score?

$$97 + 73 + 84 + 86 + 95 + 88 + 97 + 63 + 100 = 783$$

Add all the values.

Divide the sum
by the number
of values.

783

÷

9

Mean = 87



What is the MEDIAN?



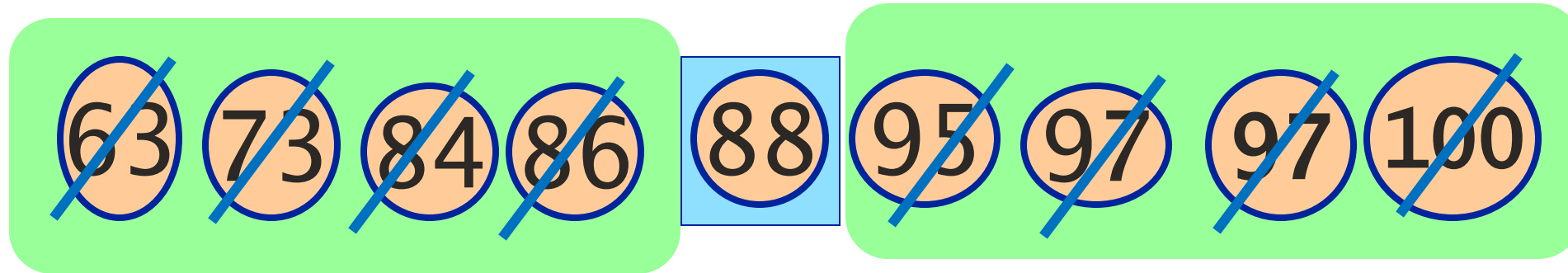
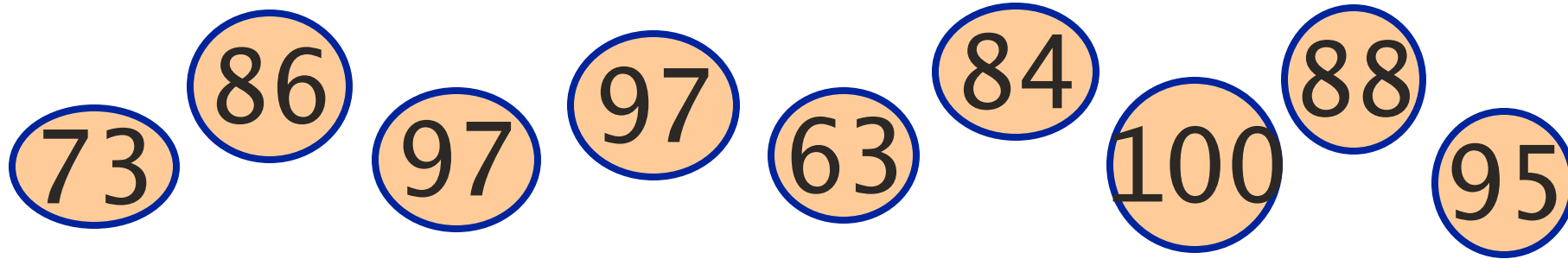
The **median** is the number that is in the **middle** of a set of data.

Steps:

- 1. Arrange the numbers in the set in order from **least to greatest**.*
- 2. Then find the number that is in the middle.*



Example: Find the median of Maha's test scores



median = 88

*Half of the numbers are
less than the median.*

*The other half are
greater than the median.*

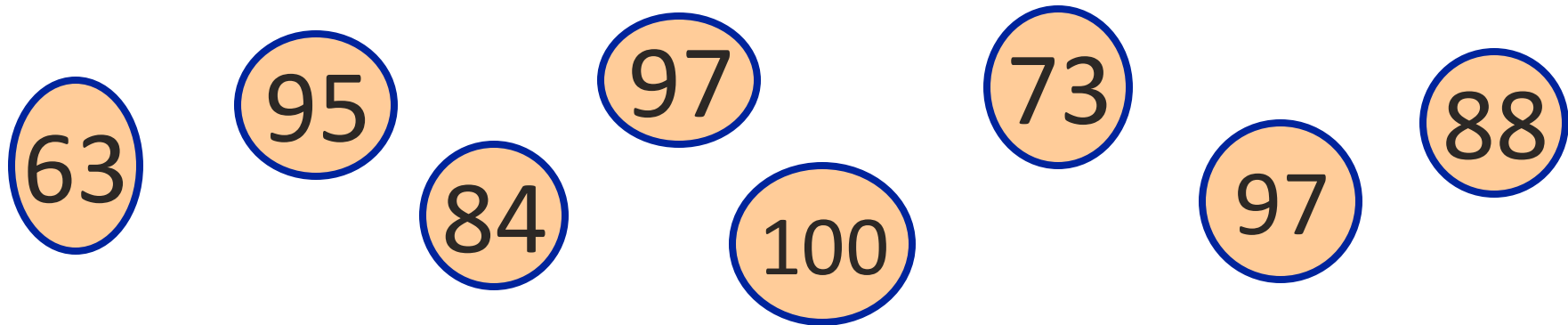


To find the **MEDIAN** for even numbers of data: (when two numbers are in the middle)

STEPS:

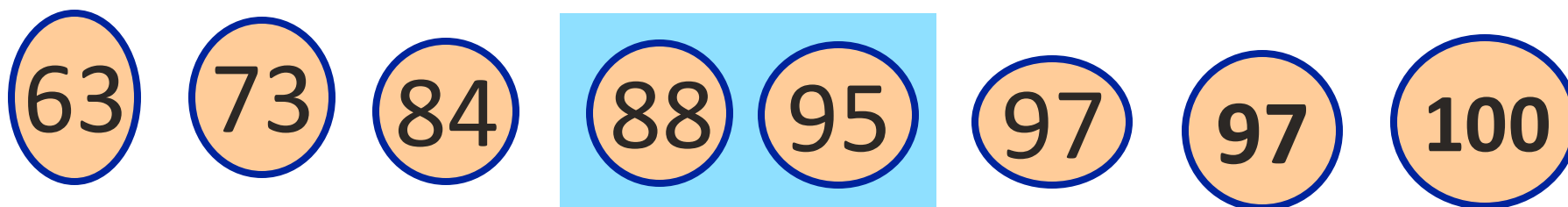
Add the two numbers.

Then divide by 2.





To find the MEDIAN for even numbers of data:
(when two numbers are in the middle)



$$88 + 95 = 183 / 2 = 91.5$$

The median is **91.5**

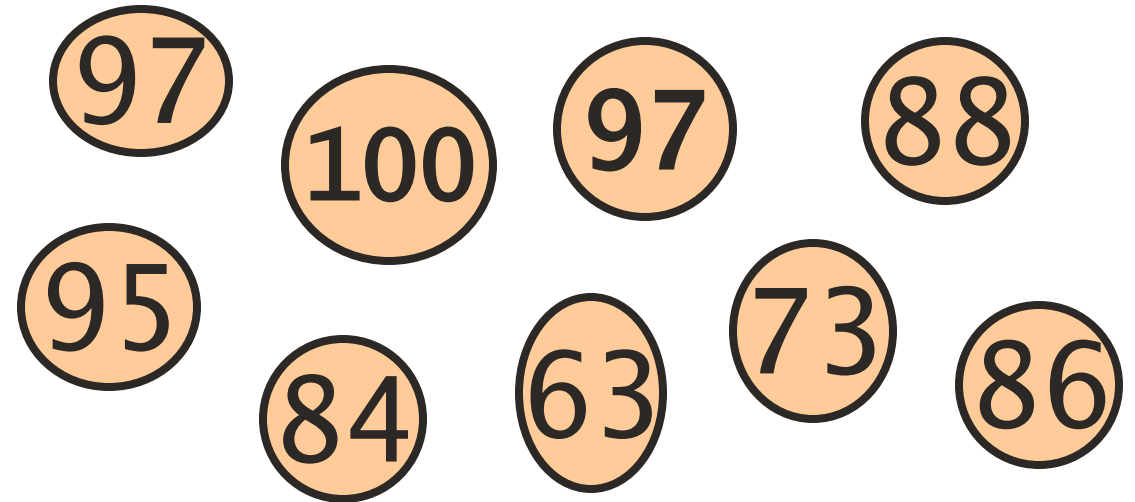


What is the MODE?

The MODE is the **piece of data that occurs most frequently in the data set.**

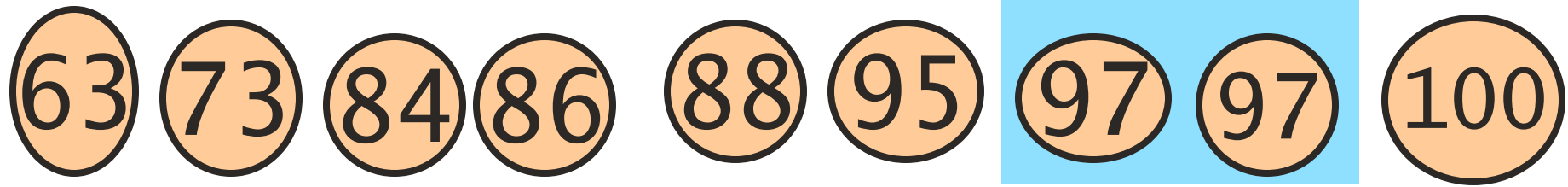
A set of data can have:

- *One mode*
- *More than one mode*
- *No mode*





What is the MODE?



The value **97** appears **twice**.
All other numbers appear just once.

MODE = 97



A hint for remembering the **MODE...**

MODE = **M**ost **O**ften



Practice 1: Which set of data has **ONE MODE**?

A

9, 11, 16, 6, 7, 17, 18

B

18, 7, 10, 7, 18

C

9, 11, 16, 8, 16



Practice 2: Which set of data has **NO MODE**?

A

9, 11, 16, 6, 7, 17, 18

B

18, 7, 10, 7, 18

C

13, 12, 12, 11, 12



Practice 3: Which set of data has **MORE THAN ONE MODE**?



A

9, 11, 16, 8, 16

B

9, 11, 16, 6, 7, 17, 18

C

18, 7, 10, 7, 18



SUMMARY:



Mean

This one is the one which requires more work than the others.



Median

*Right in the **MIDDLE**.*



Mode

*This one is the easiest to find — Just **LOOK**.*



Find the....



Mean



Median



Mode

9, 10, 10, 13, 13



Answers:



Mean

11



Median

10



Mode

10, 13



Find the....



Mean



Median



Mode

8, 8, 9, 10, 10, 12, 12, 13, 17



Answers:

 Mean

11

 Median

10

 Mode

8, 10, 12



Find the....



Mean



Median



Mode

13, 82, 79, 54, 60, 48



Answers:



Mean

56



Median

57



Mode

No mode



How to find Mean for UnGrouped Frequency Table?



Example: Parking Spaces per House in Hampton Street

Isabella went up and down the street to find out how many parking spaces each house has. Here are her results:

Parking Spaces	Frequency
1	15
2	27
3	8
4	5

What is the mean number of Parking Spaces?

Answer:

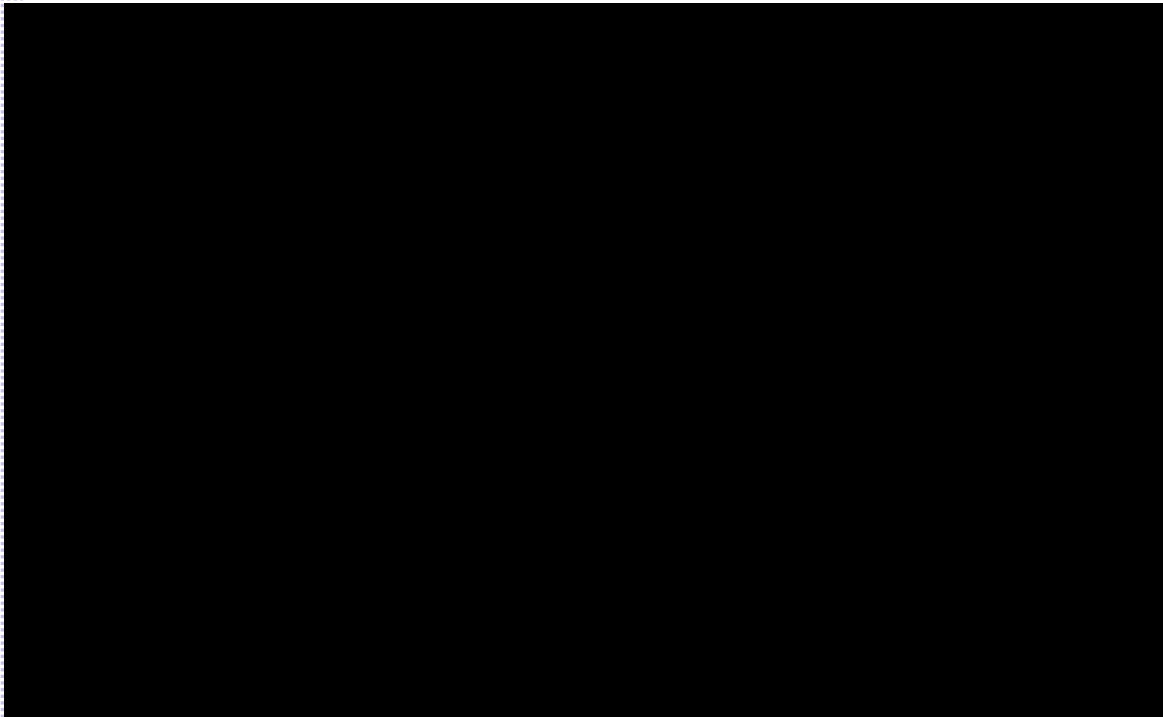
$$\begin{aligned}\text{Mean} &= \frac{15 \times 1 + 27 \times 2 + 8 \times 3 + 5 \times 4}{15 + 27 + 8 + 5} \\ &= \frac{15 + 54 + 24 + 20}{55} \\ &= 2.05\ldots\end{aligned}$$

The Mean is **2.05** (to 2 decimal places)



How to find **Mean** for **Grouped Data**?

25



Step 1: **MODE** **3** Stat **1** 1-Var

Step 2: **SHIFT** **MODE** Click on the down arrow  Select **4** Stat
Frequency? **1** On

Step 3: Enter values. Press **=** to go the next place and right arrow  to go to next column

Step 4: Press **AC** - this stores all your values in the calculator

For Mean

SHIFT **1** **4** Var

Select **2** $\bar{x}$

Press **=** This gives you mean

*To clear memory and
reset calculator*

SHIFT **9** **3**



Example 1: The data shows the maximum wind speeds for 40 states.
Find mean.



Class Limits	Midpoints	Frequency
67 - 78	72.5	3
79 - 90	84.5	5
91 - 102	96.5	8
103 - 114	108.5	9
115 - 126	120.5	5
		Total = 30

$$\text{Midpoint} = \frac{x_1 + x_2}{2}$$

Calculator Guide:

Step 1: **MODE** **3** Stat **1** 1-Var

Step 2: **SHIFT** **MODE** Click on the down arrow  Select **4** Stat
Frequency? **1** On

Step 3: Enter values. Press **=** to go the next place and right arrow  to go to next column

Step 4: Press **AC** - this stores all your values in the calculator

Mean = 99.7

To clear memory and
reset calculator

SHIFT **9** **3**



Example 2: The data shows the number of points the winning team scored in Rose Bowl. Find mean. Round your answer to 2 d.p

Class Limits	Midpoints	Frequency
17 - 20	18.5	7
21 - 24	22.5	4
25 - 28	26.5	6
29 - 32	30.5	11
33 - 36	34.5	9
		Total = 37

Calculator Guide:

Step 1: **MODE** **3** Stat **1** 1-Var

Step 2: **SHIFT** **MODE** Click on the down arrow  Select **4** Stat
Frequency? **1** On

Step 3: Enter values. Press **=** to go the next place and right arrow  to go to next column

Step 4: Press **AC** - this stores all your values in the calculator

$$\text{Midpoint} = \frac{x1 + x2}{2}$$

Mean = 27.69

To clear memory and
reset calculator

SHIFT **9** **3**



Weighted Mean

Find the **weighted mean** of a variable X by multiplying each value by its corresponding weight and dividing the sum of the products by the sum of the weights.

$$\bar{X} = \frac{w_1X_1 + w_2X_2 + \cdots + w_nX_n}{w_1 + w_2 + \cdots + w_n} = \frac{\sum wX}{\sum w}$$

where $w_1, w_2, \dots, w_n$ are the weights and $X_1, X_2, \dots, X_n$ are the values.



Word Problems:



1. Find the **weighted mean** price of three models of automobiles sold. The number and price of each model sold are given below. Round the answer to 2 d.p.

Model	Number	Price
A	8	\$10,000
B	10	12,000
C	12	8,000

$$\text{MEAN} = \frac{\text{TOTAL VALUE IN THE SET}}{\text{NUMBER OF VALUES}}$$

$$\text{Weighted Mean} = \frac{\sum wx}{\sum w}$$

$$= \frac{(8*10000)+(10*12000)+(12*8000)}{8+10+12} = \$ 9,866.67$$



Word Problems:

2. An instructor grades exam 20%, term paper 30%, final exam 50%. A student has grades 83, 72 and 90. Find student's **final average**?

$$\begin{aligned}\text{Weighted Mean} &= \frac{\sum wx}{\sum w} \\ &= \frac{(20*83) + (30*72) + (50*90)}{20+30+50} \\ &= 83.2\end{aligned}$$



Word Problems involving Mean:

1. Hussein scored 82, 79, 74 and 85 marks in the four progress tests of the first semester. What was his **average mark** for the semester?

$$\text{MEAN} = \frac{\text{TOTAL VALUE IN THE SET}}{\text{NUMBER OF VALUES}}$$

$$\text{Mean} = \frac{82+79+74+85}{4} = 80$$



Word Problems involving Mean:

2. The **average cost** of four different brands of tea is AED70 per kilogram. The prices of three of the brands are AED66, AED74 and AED76 per kilogram. What is the cost of the **fourth brand**?

$$\text{MEAN} = \frac{\text{TOTAL VALUE IN THE SET}}{\text{NUMBER OF VALUES}}$$

$$= \frac{66 + 74 + 76 + a}{4} = 70$$

$$a = 64 \text{ AED}$$



Word Problems involving Mean:

3. The average age of the German football team (11 players) is 31 years. One player aged 44 is **replaced** by a young one aged 22. What is the average age of the team now?

$$31 = \frac{\sum x(\text{old})}{11}$$

$$\sum x(\text{old}) = 31 \times 11 = 341$$

$$\text{New Mean} = \frac{\sum x(\text{old}) - 44 + 22}{11}$$

$$= \frac{341 - 44 + 22}{11}$$

$$= 29$$



Word Problems involving Mean:

4. The average age of a student at ADWC is 17 years and that at ADMC is 19 years. If ADWC has 900 students and ADMC has 1100, find the average age of the students at both the colleges.

Mean(ADWC)

$$17 = \frac{\sum x(ADWC)}{900}$$

$$\sum x(ADWC) = 17 \times 900 = 15300$$

Mean(ADMC)

$$19 = \frac{\sum x(ADMC)}{1100}$$

$$\sum x(ADMC) = 19 \times 1100 = 20900$$

$$\text{Average Mean} = \frac{\sum x(ADWC) + \sum x(ADMC)}{900 + 1100}$$

$$= \frac{15300 + 20900}{2000}$$

$$= 18.1$$



How to find Modal Class?

Example 1: Find modal class of the below data.

Class Limits	Frequency
67 - 78	3
79 - 90	5
91 - 102	8
103 - 114	9
115 - 126	5

Modal class = class
with highest
frequency

Modal Class = 103 - 114



How to find Modal Class?

Example 2: Find modal class of the below data

Class Limits	Frequency
17 - 20	7
21 - 24	4
25 - 28	6
29 - 32	11
33 - 36	9

Modal class = class
with highest
frequency

Modal Class = 29 - 32



Steps for Finding Median Class

Statistics

Step 1: Find the cumulative frequency.

Step 2: Find the position of median.

case 1: If $n = \text{even}$ $\frac{n}{2}$ *is the position*

case 2: If $n = \text{odd}$ $\frac{n+1}{2}$ *is the position*

Step 3: Get the “x” value which is in the right position that will be the median.



Example 1: The data shows the maximum wind speeds for 40 states.
Find **median class**.

Class Limits	Frequency	Cumulative Frequency
67 - 78	3	3
79 - 90	5	8
91 - 102	8	16
103 - 114	9	25
115 - 126	5	30
Total = 30		

$$\frac{n}{2} = \frac{30}{2} = 15^{\text{th}} \text{ position}$$

15th position is between 8th and 16th position. Hence, we will take the class in 16th position

Median Class = 91 - 102



Example 2: The data shows the number of points the winning team scored in

Rose Bowl. Find **median class**.

Class Limits	Frequency	Cumulative
17 - 20	7	7
21 - 24	4	11
25 - 28	6	17
29 - 32	11	28
33 - 36	9	37

Total = 37

$$\frac{n+1}{2} = \frac{37+1}{2} = \frac{38}{2} = 19^{\text{th}} \text{ position}$$

19th position is between 17th and 28th position. Hence, we will take the class in 28th position

Median Class = **29 - 32**



Summary:

TABLE 3 – 1 Summary of Measures of Central Tendency

Measure	Definition	Symbol(s)
Mean	Sum of values, divided by total number of values	$\mu, \bar{X}$
Median	Middle point in data set that has been ordered	MD
Mode	Most frequent data value	None

Properties and Uses of Central Tendency

The Mean

1. The mean is found by using all the values of the data.
2. The mean varies less than the median or mode when samples are taken from the same population and all three measures are computed for these samples.
3. The mean is used in computing other statistics, such as the variance.
4. The mean for the data set is unique and not necessarily one of the data values.
5. The mean cannot be computed for the data in a frequency distribution that has an open-ended class.
6. The mean is affected by extremely high or low values, called outliers, and may not be the appropriate average to use in these situations.



The Median

1. The median is used to find the center or middle value of a data set.
2. The median is used when it is necessary to find out whether the data values fall into the upper half or lower half of the distribution.
3. The median is used for an open-ended distribution.
4. The median is affected less than the mean by extremely high or extremely low values.

The Mode

1. The mode is used when the most typical case is desired.
2. The mode is the easiest average to compute.
3. The mode can be used when the data are nominal or categorical, such as religious preference, gender, or political affiliation.